

EXPERIMENT 19

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Problem Statement

Prepare test cases for the given SGPA/CGPA algorithm, covering valid, invalid and boundary inputs; write the corresponding C program; compile and execute it; and test it against the prepared cases.

Solution Approach

The program validates the global ID, number of semesters, number of courses, course code, credits and grade. Grade points are mapped as O=10, A=9, B=8, C=7, D=6, E=5 and P=4. For each semester, SGPA = semester_TGC / semester_TC. After all semesters, CGPA = cumulative_TGC / cumulative_TC.

Test-Case Table

TC	Input / Condition	Expected Result
1	global_id = 101; m=1; n=2; courses: (11,4,O), (12,3,A)	SGPA = CGPA = (40+27)/7 = 9.57
2	global_id = 0, then 101	Reject 0; accept 101
3	m = -1, then 1	Reject -1; accept 1
4	n = 0, then 1	Reject 0; accept 1
5	course code = -5, then 10	Reject -5; accept 10
6	credit = 0, then 4	Reject 0; accept 4
7	grade = X, then B	Reject X; accept B
8	Boundary grade O, credit=1	GP=10; semester contribution=10
9	Boundary grade P, credit=1	GP=4; semester contribution=4
10	2 semesters: S1 (3,A); S2 (4,B)	S1=9.00, S2=8.00, CGPA=(27+32)/7=8.43

C Program

```
#include <stdio.h>
#include <ctype.h>

int main(void) {
    int global_id, m, n, semester, course, code, credit, GP;
    int semester_TC, semester_TGC, cumulative_TC = 0, cumulative_TGC = 0;
    char grade;
    float SGPA, CGPA;

    do {
        printf("Enter positive global ID: ");
        scanf("%d", &global_id);
        if (global_id <= 0) printf("Invalid global ID. Try again.\n");
    } while (global_id <= 0);

    do {
        printf("Enter number of semesters: ");
        scanf("%d", &m);
        if (m <= 0) printf("Invalid number of semesters. Try again.\n");
    } while (m <= 0);

    for (semester = 1; semester <= m; semester++) {
        semester_TC = 0;
        semester_TGC = 0;

        do {
            printf("\nSemester %d - enter number of courses: ", semester);
            scanf("%d", &n);
```

```

        if (n <= 0) printf("Invalid number of courses. Try again.\n");
    } while (n <= 0);

    for (course = 1; course <= n; course++) {
        do {
            printf("Course %d code (positive integer): ", course);
            scanf("%d", &code);
            if (code <= 0) printf("Invalid course code. Try again.\n");
        } while (code <= 0);

        do {
            printf("Credits for course %d: ", course);
            scanf("%d", &credit);
            if (credit <= 0) printf("Invalid credit. Try again.\n");
        } while (credit <= 0);

        do {
            printf("Grade (O/A/B/C/D/E/P): ");
            scanf(" %c", &grade);
            grade = (char)toupper((unsigned char)grade);
            switch (grade) {
                case 'O': GP = 10; break;
                case 'A': GP = 9; break;
                case 'B': GP = 8; break;
                case 'C': GP = 7; break;
                case 'D': GP = 6; break;
                case 'E': GP = 5; break;
                case 'P': GP = 4; break;
                default:
                    GP = -1;
                    printf("Invalid grade. Try again.\n");
            }
        } while (GP == -1);

        semester_TGC += GP * credit;
        semester_TC += credit;
    }

    SGPA = (float)semester_TGC / semester_TC;
    printf("\nSemester %d SGPA = %.2f\n", semester, SGPA);

    cumulative_TGC += semester_TGC;
    cumulative_TC += semester_TC;
}

CGPA = (float)cumulative_TGC / cumulative_TC;
printf("\nGlobal ID: %d\n", global_id);
printf("Cumulative TC: %d\n", cumulative_TC);

printf("Cumulative TGC: %d\n", cumulative_TGC);
printf("CGPA = %.2f\n", CGPA);
return 0;
}

```

Sample Execution

For global ID 101, one semester and two courses:

Course 1: code 11, credit 4, grade O -> $10 \times 4 = 40$

Course 2: code 12, credit 3, grade A -> $9 \times 3 = 27$

Semester TC = 7, Semester TGC = 67

SGPA = $67 / 7 = \mathbf{9.57}$

Since there is only one semester, CGPA = **9.57**.

Result

The C program implements the supplied algorithm, validates invalid inputs using repeated prompts, calculates semester-wise SGPA and cumulative CGPA, and covers valid, invalid and boundary test cases.